

iQ-Guard: An Effective and Noise-Resistant Framework for Graph Fraud Detection on iQIYI Platform

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Abstract

Fraud detection is a critical challenge for large-scale online platforms, where even a small fraction of fraudulent users can cause millions of dollars in financial losses. In practice, graph structures are commonly adopted to model interactions, such as transactions and user behaviors. However, fraudsters deliberately camouflage their behaviors to mimic legitimate users, introducing both label noise and structural noise into the graph-based models. To this end, we develop iQ-Guard, an effective and noise-resistant framework for graph fraud detection that has been fully deployed on iQIYI, one of China's largest online video and social platforms.

iQ-Guard proposes two novel components: (i) Dynamic Contrastive Learning, which adaptively augments data for each sample, enabling the model to resist noisy labels while preserving meaningful representation information gradually, and (ii) Structure-aware Noise-resistant Training, which leverages edge relations and robust loss functions to mitigate both label and structural noise. Extensive experiments on five datasets, including large-scale industrial data, show that iQ-Guard outperforms 14 state-of-the-art methods. Since its deployment, iQ-Guard has detected an average of 36 fraud cases per day, preventing over \$1.12 million in financial losses.

CCS Concepts

• Computing methodologies → Neural networks; • Security and privacy → Web application security.



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Keywords

Fraud Detection; Robust Learning; Data Augmentation

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1 Introduction

Detecting sophisticated fraud within large-scale relational data is a critical challenge for online platforms. For example, iQIYI, one of China's largest video platforms, serves over 30 million daily active users, among which approximately 0.28% of accounts exhibit abnormal or fraudulent behaviors. Although this ratio seems small, it corresponds to tens of thousands of suspicious accounts each day, posing significant risks of financial loss and reduced user trust. In production environments, rule-based strategies relying on predefined heuristics typically miss more than 20% of fraudulent cases while producing numerous false alarms. Such misjudgments not only increase manual review costs but also result in annual financial losses of millions of dollars.

Graph Fraud Detection (GFD) aims to identify fraudulent nodes in networks by analyzing hidden relationships and patterns indicative of fraud [32, 48, 53]. It has broad applications in finance [51], telecommunications [64], and social networks [32]. Existing GFD approaches are generally divided into two categories: (i) **spatial-based methods**, which focus on message passing and feature propagation across graphs [21, 34, 40, 56, 63, 67], and (ii)

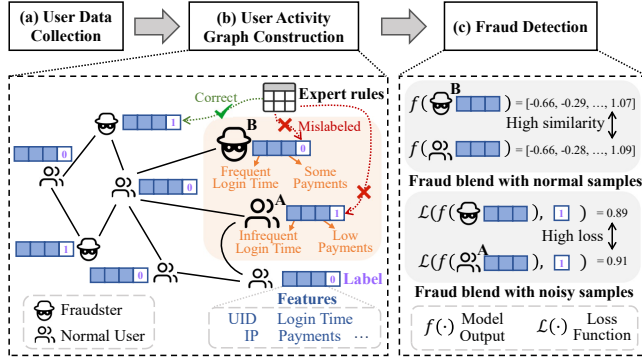


Figure 1: Misjudgments in expert rules lead to the misclassification of both normal user A and fraudster B.

spectral-based methods, which employ spectral filters to distinguish fraudsters from legitimate users [3, 19, 45, 54, 61].

However, GFD in industrial settings is often limited by the entanglement of **label noise** and **structural noise**. Here, we give a concrete example that arises in iQIYI’s incentive system, where users receive rewards (e.g., coupons) for engagement.

As shown in Fig. 1, the platform employs a three-stage fraud detection pipeline: **(a) User Data Collection**. The platform collects user activity logs, including viewing, purchasing, and engagement behaviors. **(b) User Activity Graph Construction**. Each user is represented as a node in a graph, with edges encoding behavioral or relational connections (e.g., shared devices, friendships). Features are derived from users’ historical activities, while labels are generated by expert rules (e.g., anomalous payments or usage thresholds). This process inevitably introduces mislabeling—e.g., a legitimate user A may be incorrectly flagged as fraudulent due to low activity, while a fraudster B mimicking normal behavior may be mislabeled as benign. **(c) Fraud Detection**. A graph-based model is then trained on this noisy graph. Fraudsters such as B deliberately mimic benign embeddings, leading to **high similarity** between fraudulent and normal nodes. At the same time, mislabeled users such as A, when mixed with true fraudsters, produce **high losses**, further confusing the model. These confounding effects severely hinder the ability to distinguish genuine fraud from noise, especially under low supervision or high-noise conditions.

The above scenario reveals **two unaddressed challenges**:

- **Label Noise**. Mislabeled benign users and truly fraudulent cases often incur similarly high training losses. This undermines the small-loss principle by making it difficult to discern whether a high loss comes from a mislabeled benign user or a genuinely complex fraudulent case, thereby hindering noise correction and degrading detection performance.
- **Structural Noise**. Fraudsters deliberately introduce heterophilic connections by linking with benign users, generating structural noise [21]. Through message passing in GNNs, these connections propagate benign features to fraudulent nodes, camouflaging them and weakening the model’s discriminative ability.

Existing approaches to label-noise robustness often rely on feature dissimilarity, an assumption that breaks down when fraudsters deliberately camouflage their behaviors. Conversely, structural-noise mitigation methods usually assume clean labels, which is

rarely the case in practice. Ultimately, prior work overlooks the concurrent and correlated nature of label and structural noise, leaving a gap between academic research and industrial needs.

To address these challenges, we propose **iQ-Guard**, a novel framework for effective and noise-resistant graph fraud detection. iQ-Guard operates in two phases:

Phase I: Dynamic Contrastive Learning (DCL). This phase focuses on improving the model’s generalization and performance by training a robust feature encoder with DCL. The core strategy leverages data augmentation as a training mechanism to help the contrastive learning capture essential, noise-invariant features.

Phase II: Structure-aware Noise-resistant Training (SNT). iQ-Guard addresses camouflage tactics by reframing fraud detection as an edge classification problem, capturing structural relations between edges. Besides, we introduce a robust loss function optimized with soft logical XOR, forming a structure-aware noise-resistant training mechanism to mitigate both structural and label noises.

Through these two phases, iQ-Guard increases the representational distance between normal users and fraudsters, reduces misleading similarities, and isolates noisy samples, ensuring that high-loss instances do not compromise model performance. Overall, the key contributions of this paper are summarized as follows:

- We propose iQ-Guard, an effective and noise-resilient framework for graph fraud detection that addresses the challenges of label and structural noise in real-world settings.
- To address label noise, iQ-Guard uses Dynamic Contrastive Learning, where augmentation strength adapts to label confidence, guiding the encoder to learn noise-invariant representations and enhancing generalization.
- To address structural noise, iQ-Guard develops structure-aware, noise-resistant training. By transforming the structural relations of edges and optimizing from the perspective of soft logical XOR, we further enhance the model’s ability to detect structural noise and mitigate label noise.
- Extensive experiments on five real-world datasets demonstrate that iQ-Guard not only outperforms 14 state-of-the-art graph fraud detection methods but also maintains robustness under label noise. Furthermore, ablation studies with alternative general designs confirm that the two key modules—Dynamic Contrastive Learning and Structure-aware Noise-resistant Training—are essential to iQ-Guard’s superior performance.
- iQ-Guard has been deployed on iQIYI, one of the largest online video-sharing and social media platforms in China, for daily fraud detection, supporting the monitoring of 30 million users. It captures an average of 36 fraud cases per day, preventing a cumulative financial loss of \$1.12 million.

2 Preliminary

We proceed to present the concept of a multi-relational graph, followed by an introduction to the general task of graph fraud detection, and then formulate our task.

2.1 Multi-relational Graph

The entities and relationships are modeled as a **multi-relational graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{Y})$, composed of (i) a set of N nodes $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ ($N = |\mathcal{V}|$), (ii) adjacency matrices of corresponding relations $\mathcal{E} = \{A_1, A_2, \dots, A_R\}$ ($R = |\mathcal{E}|$), (iii) a set of node

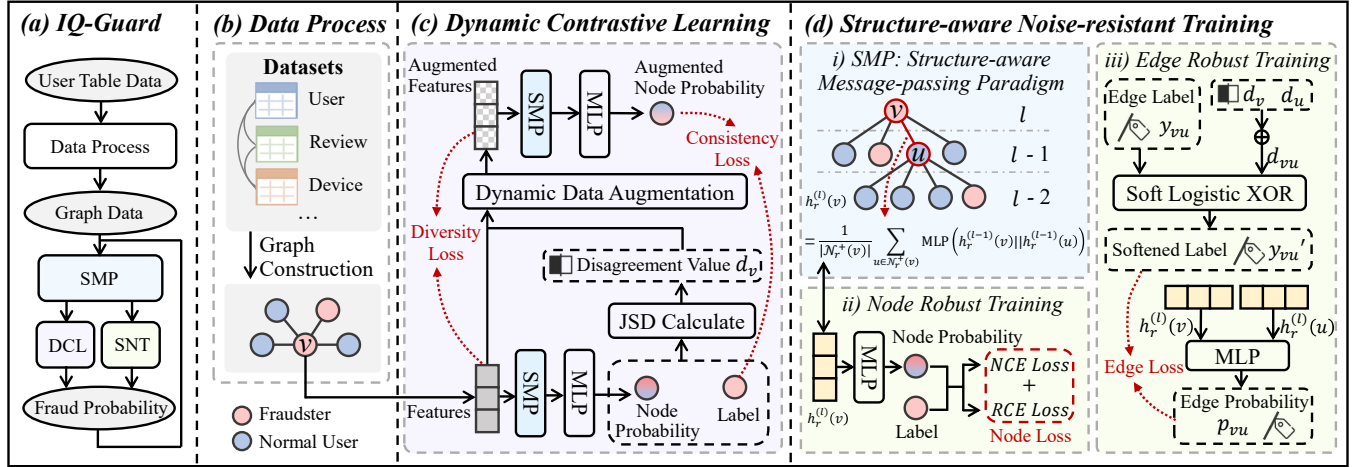


Figure 2: The framework of iQ-Guard.

feature vectors $\mathcal{X} = \{x_1, x_2, \dots, x_N\}$, and (iv) a node label set $\mathcal{Y} = \{y_1, y_2, \dots, y_N\}$. Nodes v and u are connected under relation $r \in \{1, 2, \dots, R\}$ if $A_r[v, u] = 1$. Each node $v \in \mathcal{V}$ has a n -dimensional feature vector $x_v \in \mathbb{R}^n$ and a label $y_v \in \{0, 1\}$, where 1 represents fraud nodes and 0 represents normal nodes.

In Fig. 1, the nodes $\mathcal{V}$ represent users, the adjacency matrices $\mathcal{E}$ capture various user interactions, the node features $\mathcal{X}$ encode users' historical behaviors, and the node labels $\mathcal{Y}$ constitute a pseudo-label set generated by expert rules.

2.2 Graph Fraud Detection

Given a graph $\mathcal{G}$ and considering a semi-supervised scenario where only a small proportion of nodes $\mathcal{V}$ have been labeled and the node label set $\mathcal{Y}$ contains noise, **graph fraud detection** is to identify whether the remaining unlabeled nodes are fraudulent.

3 Methodology

The framework of iQ-Guard is illustrated in Fig. 2. It mitigates the adverse effects of both structural and label noise through two key modules: (i) Dynamic Contrastive Learning (DCL) and (ii) Structure-aware Noise-resistant Training (SNT).

In the data process phase, raw user activity data is transformed into a graph representation, which then serves as the input for these modules. In DCL module, the Jensen–Shannon Divergence (JSD) of each sample is computed from the model's previous predictions, and the augmentation strength is dynamically adjusted in each iteration. High-confidence samples receive stronger perturbations, guiding the model to capture invariant features and thus improving robustness. In SNT module, a composite objective is introduced. For nodes, a robust classification loss combining Normalized and Reverse Cross-Entropy reduces sensitivity to noisy labels. For edges, uncertainty-aware labels are generated based on JSD scores of connected nodes, refining structural learning. The overall model is trained by jointly optimizing the node loss, the edge loss, and an auxiliary consistency term.

3.1 Dynamic Contrastive Learning

Design Motivation. Contrastive learning (CL) has been shown to enhance robustness against label noise [10, 55, 62]. However, most existing CL methods treat all labeled samples uniformly [29, 35],

overlooking the fact that not all labels are noisy. This indiscriminate treatment often leads to either over-processing clean labels or under-processing corrupted ones, thereby weakening supervision. To address this challenge, we propose a Dynamic Contrastive Learning (DCL) framework that is inherently designed for large-scale noisy datasets. This design not only improves robustness to label noise but also ensures scalability and efficiency in real-world industrial settings, where adaptive handling of data quality is critical for reliable deployment.

3.1.1 Label Confidence Computation. Given a training graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{Y})$, we first input it into a graph neural network (GNN) θ with structure-aware message-passing paradigm to estimate label confidences, which allows us to distinguish reliable samples from potentially corrupted ones. Specifically, the model produces a probability prediction p_v for each node $v \in \mathcal{V}$. The disagreement between prediction p_v and ground-truth label y_v is quantified using Jensen–Shannon divergence (JSD):

$$d_v = \frac{1}{2} \text{KL} \left(y_v \parallel \frac{y_v + p_v}{2} \right) + \frac{1}{2} \text{KL} \left(p_v \parallel \frac{y_v + p_v}{2} \right), \quad (1)$$

where $d_v \in [0, 1]$ measures the degree of label disagreement and $\text{KL}(\cdot)$ represents the Kullback-Leibler divergence [9].

While prior works often utilize cross-entropy loss values to fit a Gaussian Mixture Model (GMM) for partitioning samples [22, 33], we adopt JSD due to its symmetric nature and inherent boundedness, which provides a stable and normalized metric for disagreement. This computation yields a set of disagreement scores $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ ($N = |\mathcal{V}|$), where a lower d_v implies greater confidence in label correctness, while higher d_v indicates potential mislabeling. This step explicitly distinguishes trustworthy from noisy labels, providing a principled confidence signal to guide subsequent augmentation and loss design.

3.1.2 Dynamic Data Augmentation. For data augmentation, we employ a noise injection strategy, a technique proven effective for graph-structured data [7, 57]. Specifically, we perturb the embedding x_v of each node v by adding a dynamically scaled noise vector:

$$x'_v = x_v + \Delta_v, \quad (2)$$

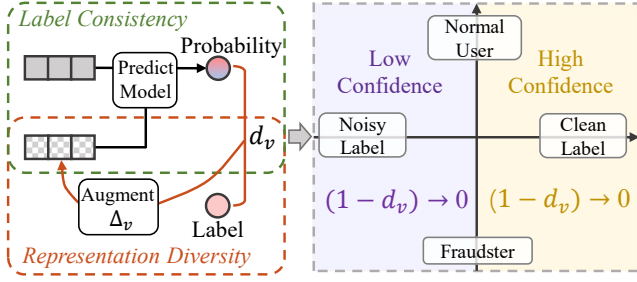


Figure 3: Label consistency, representation diversity, and sample confidence in DCL.

where the noise vector Δ_v is defined as $\Delta_v = \delta \odot \text{sgn}(x_v)$ with $\|\Delta_v\|_2 = \varepsilon$. The parameter ε controls the perturbation radius, ensuring that augmented embeddings remain close to the original representations.

To tie the augmentation strength to our derived confidence, the elements of δ are sampled from a distribution whose variance is a function of the disagreement score d_v . We propose sampling from a scaled uniform distribution:

$$\delta \sim U(0, 1 - d_v), \quad (3)$$

where nodes with high label confidence (low d_v) are injected with stronger perturbations, encouraging the model to learn more stable and noise-invariant representations from reliable supervision. Conversely, nodes with low confidence receive milder perturbations, avoiding reinforcement of potentially corrupted labels. By coupling augmentation strength with label confidence, the model enforces robustness against noise while preserving clean-label information.

3.1.3 Contrastive Loss Function. To guide the model toward learning robust and discriminative representations, particularly in the presence of noisy labels, we formulate a contrastive learning (CL) layer ϕ , which is composed of two complementary components: a **label consistency loss** and a **representation diversity loss**. As shown in Fig. 3, both components are adaptively weighted by $(1 - d_v)$, which reflects the confidence level of sample v . In this way, high-confidence samples contribute more strongly to the optimization, while low-confidence samples are down-weighted to mitigate the effect of noisy labels.

First, we introduce the **label consistency** component to ensure that the semantic meaning of a node remains invariant under augmentation. Specifically, an augmented sample x'_v should share the same label as its original counterpart x_v . Let y_v be the ground-truth label of the original node and p'_v be the predicted probability distribution for the augmented node. We enforce this consistency using a confidence-weighted cross-entropy objective:

$$\mathcal{L}_{\text{cons}} = - \sum_{v \in \mathcal{V}} (1 - d_v) y_v \cdot \ln p'_v, \quad (4)$$

where $(1 - d_v)$ acts as a confidence weight, amplifying the contribution of reliable samples while suppressing the gradient from potentially mislabeled ones.

Second, to discourage the model from learning a trivial mapping where the augmented representation collapses to the original, we introduce a **representation diversity** component. This objective encourages the augmented node embedding x'_v to be distinct from the original embedding x_v in the representation space. We promote

this diversity by maximizing a distance metric $q(\cdot, \cdot)$ between the two representations, equivalently minimizing the negative distance:

$$\mathcal{L}_{\text{div}} = - \sum_{v \in \mathcal{V}} (1 - d_v) q(x_v, x'_v), \quad (5)$$

where $q(\cdot, \cdot)$ denotes a selected distance metric, such as Euclidean distance or cosine similarity. The confidence weight $(1 - d_v)$ again ensures that diversity is primarily enforced on trustworthy samples.

The final contrastive learning objective is a weighted combination of the consistency and diversity losses:

$$\mathcal{L}_{\text{CL}} = \mathcal{L}_{\text{cons}} + \mathcal{L}_{\text{div}}, \quad (6)$$

where the hyperparameter α controls the trade-off between maintaining semantic consistency and promoting informative diversity. This composite loss function guides the model to learn discriminative node embeddings that are robust to label noise by balancing semantic invariance with representational diversity.

3.2 Structure-aware Noise-resistant Training

Design Motivation. Camouflage tactics essentially introduce structural noise, disrupting the original graph structure. Existing methods attempt to repair structural noise using contextual label information [6, 50, 76], but in real-world scenarios, label noise is inevitable, which may further mislead the model. This challenge becomes even more critical at industrial scale, where noise can accumulate and propagate through massive graphs. To address this, iQ-Guard introduces noise-resistant training grounded in structural information rather than relying solely on labels. With dual resistance to structural and label noise, iQ-Guard delivers scalable robustness for large-scale industrial graph deployment.

3.2.1 Structure-aware Message-passing Paradigm. We incorporate edge-structural information by representing each relation-specific edge (u, v, r) with the concatenation of its endpoint features. Let $\mathcal{N}_r(v) = \{u \in \mathcal{V} \mid A_r[v, u] = 1\}$ and $\mathcal{N}_r^+(v) = \mathcal{N}_r(v) \cup \{v\}$. We initialize $h_r^{(0)}(v) = x_v$.

For $l \geq 1$, the structure-aware message passing is

$$h_r^{(l)}(v) = \frac{1}{|\mathcal{N}_r^+(v)|} \sum_{u \in \mathcal{N}_r^+(v)} \text{MLP}_{\psi_r} \left([h_r^{(l-1)}(v) \parallel h_r^{(l-1)}(u)] \right), \quad (7)$$

where $[\cdot \parallel \cdot]$ denotes vector concatenation, and MLP_{ψ_r} is a three-layer network parameterized by ψ_r , with parameters shared across all edges under relation r .

3.2.2 Node Robust Training. For the fraud detection task, we design a robust node loss function. After obtaining the final node representation $h_r^{(l)}(v)$ from the structure-aware message-passing layers, we generate a class prediction for the fraud detection task. This is accomplished by feeding the representation into a prediction head, $f(\cdot)$, which is implemented as a linear classifier to produce the final output probability:

$$p_v = f(h_r^{(l)}(v)) = \text{MLP}(h_r^{(l)}(v)). \quad (8)$$

According to existing studies [4, 20], a binary classification loss $\mathcal{L}(\cdot)$ is considered robust to label noise if the sum of losses over all possible labels remains constant, i.e., $\sum_{j \in \{0,1\}} \mathcal{L}(p_v, j) = C$, where

C is a constant. This condition ensures that noisy labels do not bias the optimization process.

However, the standard cross-entropy (CE) loss does not satisfy this property. To enforce it, we normalize CE by the sum of losses across both labels, yielding the following normalized formulation:

$$\mathcal{L}_{norm} = \frac{\mathcal{L}_{CE}(p_v, y_v)}{\sum_{j \in \{0,1\}} \mathcal{L}_{CE}(p_v, j)}. \quad (9)$$

This normalization guarantees that $\sum_{j \in \{0,1\}} \mathcal{L}_{norm}(p_v, j) = 1$, thus meeting the robustness criterion.

By substituting the binary CE expression into Equation 9, we obtain the **Normalized Cross-Entropy (NCE)** loss:

$$\mathcal{L}_{NCE} = \sum_{v \in \mathcal{V}} \frac{y_v \log p_v + (1 - y_v) \log(1 - p_v)}{\log p_v + \log(1 - p_v)}. \quad (10)$$

While NCE enhances robustness, it may also lead to underfitting. To improve discriminability, we incorporate **reverse cross-entropy (RCE)** [49], which exploits complementary supervision signals from incorrect labels:

$$\mathcal{L}_{RCE} = - \sum_{v \in \mathcal{V}} [p_v \log(y_v) + (1 - p_v) \log(1 - y_v)]. \quad (11)$$

However, RCE suffers from numerical instability when predictions approach 0 or 1, as the $\log(0)$ term causes divergence and unstable gradients. To address this, we introduce a normalized variant, denoted as NRCE, which scales the RCE term by a stabilizing constant A (empirically set to -4):

$$\mathcal{L}_{NRCE} = \sum_{v \in \mathcal{V}} \frac{p_v \log(y_v) + (1 - p_v) \log(1 - y_v)}{A}. \quad (12)$$

This normalization prevents numerical divergence while preserving the complementary learning signal, allowing NRCE to serve as a stable extension of RCE for robust fraud detection.

By jointly applying NCE, which ensures robustness, and NRCE, which provides complementary supervision, we obtain the final robust node loss:

$$\mathcal{L}_{node} = \mathcal{L}_{NCE} + \mathcal{L}_{NRCE}. \quad (13)$$

This combination balances resistance to noisy labels and the preservation of discriminative training signals.

3.2.3 Edge Robust Training. Under label noise, the supervision derived from edge labels can be misleading. Importantly, the structural information between two nodes changes only when one node is mislabeled while the other is correctly labeled—analogous to an XOR operation. To capture this property, we compute a soft logical XOR [2] based on the label confidences of two connected nodes.

For an edge (v, u) , we define the structural confidence d_{vu} as:

$$d_{vu} = d_v \oplus d_u = \min(\max(d_v - d_u, 0) + \max(d_u - d_v, 0), 1), \quad (14)$$

where d_v and d_u represent the label confidence of node v and its neighboring node u , respectively, which are calculated in Section 3.1. If both node labels are either incorrect or correct, d_{vu} is close to 0; otherwise, it tends toward 1.

To mitigate the effect of noisy supervision, we soften the edge labels:

$$y'_{vu} = \begin{cases} 1 - d_{vu}, & \text{if } y_{vu} = 1 \\ d_{vu}, & \text{if } y_{vu} = 0, \end{cases} \quad (15)$$

where $y_{vu} = \mathbb{I}(y_v, y_u)$ is the original hard edge label, and $\mathbb{I}(\cdot)$ is an indicator function, defined as $\mathbb{I}(\cdot) = 1$ if the internal condition is satisfied, and $\mathbb{I}(\cdot) = 0$ otherwise.

The robust edge classification loss is then defined as:

$$\mathcal{L}_{edge} = - \sum_{v \in \mathcal{V}} \sum_{u \in \mathcal{N}_v} y'_{vu} \ln p_{vu}, \quad (16)$$

where $p_{vu} = \sigma(\text{MLP}_{\psi_r}(h_r^{(l)}(v) \parallel h_r^{(l)}(u)))$, σ is a sigmoid function and the MLP parameters are consistent with ψ_r as mentioned above.

By integrating soft logical XOR-based structural confidence into edge supervision, our method adaptively reduces the influence of noisy labels while preserving informative structural signals, thereby enhancing robustness in edge-level classification.

3.2.4 Overall Objective. Finally, the overall loss of our model consists of three components: the contrastive loss function $\mathcal{L}_{CL}$, the edge classification loss function $\mathcal{L}_{edge}$, and the node classification loss function $\mathcal{L}_{node}$. Therefore, the total loss is computed as follows:

$$\mathcal{L}_{fraud} = \mathcal{L}_{CL} + \mathcal{L}_{edge} + \mathcal{L}_{node}. \quad (17)$$

This composite loss balances representation learning, structural supervision, and node classification, leading to a unified framework that is robust against both label and structural noise.

The overall training algorithm and corresponding complexity analysis are detailed in **Appendix B**.

4 Experiments

We evaluate iQ-Guard by the following Research Questions (RQs):

RQ1: How does iQ-Guard perform compared with baselines?

RQ2: How robust is iQ-Guard against label noise?

RQ3: What is the contribution of each module to the overall performance of iQ-Guard?

RQ4: How does iQ-Guard jointly address structural and label noise?

RQ5: What are the real-world deployment results of iQ-Guard on the iQIYI platform?

RQ6: How robust is iQ-Guard with respect to parameter sensitivity?

4.1 Experimental Setup

Datasets. We conduct experiments on four public datasets and one industrial dataset: T-Finance, T-Social, YelpChi, Amazon, and iQIYI, with detailed descriptions provided in **Appendix C.1**. As the public datasets do not contain label noise, we follow prior studies and inject 20% artificial noise using the widely adopted symmetric noise model [33, 38, 69]. In contrast, the iQIYI dataset, collected from a real-world production environment, naturally contains label noise, with an estimated noise ratio of around 15%.

Baselines. We compare iQ-Guard with 14 baseline methods, which fall into four categories. Due to space limitations, detailed descriptions of these methods are provided in **Appendix C.2**.

- **Traditional GNN Methods.** GCN [31], GAT [47], and GraphSAGE [5].
- **Noise-resistant Methods.** NRGNN [11] and GraphCleaner [36].
- **Contrastive Learning based Methods.** GraphMAE [25], and GraphACL [60].
- **Graph Fraud Detection Methods.** BWGNN [45], SparseGAD [21], GAGA [50], ConsisGAD [6], PMP [76], HOGRL [77] and RGTAN [58].

Table 1: The overall performance. The highest and second-highest scores are highlighted in bold and underlined, respectively.

Methods	Amazon			YelpChi			T-Finance			T-Social			iQIYI		
	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1	AUC	AP	F1
GCN	0.7396	0.4101	0.6147	0.5213	0.1920	0.4463	0.6164	0.4725	0.5948	0.6389	0.5064	0.4337	0.6402	0.3003	0.5214
GAT	0.7286	0.2774	0.6013	0.5416	0.2526	0.4396	0.7132	0.6148	0.5023	0.5878	0.6229	0.4582	0.6557	0.2984	0.5336
GraphSAGE	0.8438	0.7607	0.7007	0.6485	0.2997	0.5521	0.6017	0.5553	0.4929	0.6163	0.4524	0.5036	0.6431	0.3125	0.5264
NRGNN	0.8159	0.5045	0.7380	0.6903	0.3367	0.5405	0.8023	0.5639	0.7128	0.7336	0.5201	0.4594	0.6608	0.3217	0.5182
GraphCleaner	0.7815	0.5001	0.6505	0.6814	0.4909	0.7391	0.8114	0.5726	0.7205	0.7475	0.5387	0.4698	0.6515	0.3109	0.5289
GraphMAE	0.8516	0.7057	0.8194	0.7324	0.4210	0.6185	0.8376	0.5994	0.7628	0.7583	0.5292	0.5019	0.6802	0.4334	0.6698
GraphACL	0.8735	0.7398	0.8402	0.7693	0.4569	0.6467	0.8403	0.6058	0.7602	0.7694	0.5524	0.5098	0.6923	0.4172	0.6308
BWGNN	0.9220	0.7182	0.8748	0.8600	0.3926	0.7245	0.9024	0.7162	0.8231	0.8967	0.7274	0.8042	0.8386	0.7083	0.7865
SparseGAD	0.8934	0.8060	0.8423	0.8156	0.6045	0.7051	0.8435	0.6642	0.7906	0.7881	0.6021	0.5398	0.8004	0.7305	0.7613
GAGA	0.8950	0.8185	0.8494	0.8764	0.7445	0.7740	0.8663	0.6912	0.8134	0.8142	0.6293	0.5639	0.8246	0.7546	0.7225
ConsisGAD	0.8242	0.7514	0.8010	0.8198	0.6798	0.7046	0.8293	0.6552	0.7702	0.7771	0.5994	0.5202	0.8124	0.7401	0.7743
PMP	0.9010	0.8221	0.8546	0.8710	0.7342	0.7612	0.9102	0.7581	0.8432	0.9273	0.8305	0.8506	0.8531	0.7734	0.7902
HOGRL	0.9051	0.8270	0.8543	0.8832	0.7460	0.7972	0.8989	0.7367	0.8510	0.8874	0.8096	0.8282	0.8858	0.7962	0.8221
RGTAN	0.8898	0.8229	0.8561	0.8571	0.6972	0.7420	0.8542	0.6761	0.7991	0.8006	0.6179	0.5521	0.7879	0.7831	0.7858
iQ-Guard	0.9469	0.8557	0.9250	0.9126	0.7847	0.8203	0.9380	0.8063	0.8836	0.9400	0.8701	0.8729	0.9648	0.8101	0.8755
Improvement	2.70%	3.47%	5.74%	3.33%	5.19%	2.90%	3.05%	6.36%	3.83%	1.37%	4.77%	2.62%	8.92%	1.75%	6.50%

Performance Metrics. In this paper, we formulate our task as a binary classification task and use three widely-used metrics: **AUC**, **AP**, and **F1**, which are detailed in [Appendix C.3](#).

Implementation Details. The setup and implementation of all baselines and iQ-Guard are detailed in [Appendix C.4](#).

4.2 Overall Performance Study (RQ1)

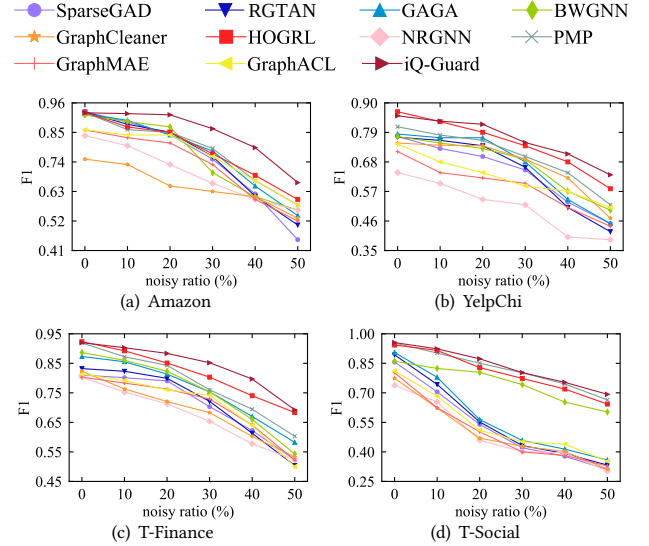
Table 1 presents the overall performance comparison on four public datasets and the iQIYI dataset.

i) iQ-Guard on public datasets. We begin with a performance comparison across four public datasets. **iQ-Guard consistently outperforms all baselines** by capturing fraudulent patterns while mitigating structural and label noise. Notably, contrastive learning and noise-resistant techniques—such as GraphCleaner and GraphACL—outperform traditional GNN models, achieving an average F1 score improvement of 15.71%. However, their performance remains substantially lower than models specifically designed for graph fraud detection methods, such as PMP and HOGRL, with F1 score deficits ranging from 10.27% to 15.08%. This suggests that existing contrastive learning and noise-resistant approaches are not fully optimized for the unique challenges of fraud detection.

ii) iQ-Guard on iQIYI dataset. As a real industrial dataset, it is characterized by large-scale data, complex relations, and inherent label noise. Nevertheless, **iQ-Guard achieves state-of-the-art performance, surpassing all baselines across all metrics.** Specifically, iQ-Guard attains an AUC of 0.9648, AP of 0.8101, and F1 score of 0.8755, representing an average improvement of 6.65%. These results highlight iQ-Guard’s strong ability to handle both structural and label noise in real-world scenarios.

4.3 Noise-Resistant Study (RQ2)

We evaluate iQ-Guard’s robustness against label noise by varying noise ratios from 0% to 50%. To ensure precise control, we introduce artificial noise into public datasets rather than relying on the naturally noisy iQIYI dataset. As shown in Fig. 4, iQ-Guard consistently outperforms all baselines, demonstrating superior resistance to noise. Specifically, its F1 score decreases by only 19.47%, significantly lower than the 34.93% drop in baselines. While some models,

**Figure 4: Performance on F1 under varying noisy ratios.**

such as GraphCleaner and GraphACL, show a more gradual decline (22.43% on average), their overall performance remains lower. These results confirm that iQ-Guard effectively mitigates label noise, enhancing fraud detection reliability in real-world scenarios.

4.4 Ablation Study (RQ3)

We evaluate iQ-Guard’s core components by modifying or removing key modules. Due to space limitations, the results on T-Finance and T-Social are provided in [Appendix C.5](#).

(i) iQ-Guard vs. iQ-Guard w/o DCL. The variant **iQ-Guard w/o DCL** removes the Dynamic Contrastive Learning, retaining only structure-aware noise-resistant training. We further replace DCL with commonly used data augmentation techniques—**Node Dropout (ND)** [15], **Edge Dropout (ED)** [43], and **Random Walk (RW)** [65]—denoted as w/o DCL (ND/ED/RW). The results show that iQ-Guard consistently outperforms all alternatives, achieving an average improvement of **5.82%** in F1. Among the alternatives,

Table 2: AUC and F1 of iQ-Guard in ablation.

Methods	Amazon		YelpChi		iQIYI	
	AUC	F1	AUC	F1	AUC	F1
iQ-Guard	0.9469	0.9250	0.9126	0.8203	0.9648	0.8755
iQ-Guard w/o DCL	0.8607	0.8325	0.8123	0.7501	0.9012	0.8253
iQ-Guard w/o DCL (ND)	0.8663	0.8469	0.8201	0.7656	0.9087	0.8380
iQ-Guard w/o DCL (ED)	0.9089	0.8793	0.8467	0.7884	0.9279	0.8605
iQ-Guard w/o DCL (RW)	0.9075	0.8782	0.8435	0.7869	0.9262	0.8583
iQ-Guard w/o SMP (GCN)	0.8723	0.8501	0.7012	0.6567	0.8743	0.8124
iQ-Guard w/o SMP (PMP)	0.9185	0.8556	0.8874	0.7603	0.9391	0.8510
iQ-Guard w/o SMP (HOGRL)	0.9276	0.8659	0.9025	0.8004	0.9463	0.8621
iQ-Guard w/o RLF	0.9005	0.8136	0.8911	0.8037	0.9228	0.8498

w/o DCL (ED) performs second best, likely because **Edge Dropout** preserves structural information more effectively. In contrast, our **dynamic data augmentation** introduces perturbations without altering the graph structure, thereby enhancing contrastive learning and validating the effectiveness of DCL.

(ii) **iQ-Guard vs. iQ-Guard w/o SMP.** The variant **iQ-Guard w/o SMP** removes the Structural-aware Message-passing Paradigm, replacing it with **GCN**, **PMP**, or **HOGRL**, denoted as w/o SMP (GCN/PMP/HOGRL). iQ-Guard surpasses all variants, achieving an average improvement of **3.61%** in F1 compared with the best substitute (HOGRL). By explicitly modeling structural information during message passing, SMP significantly improves robustness to label noise and contributes substantially to overall performance.

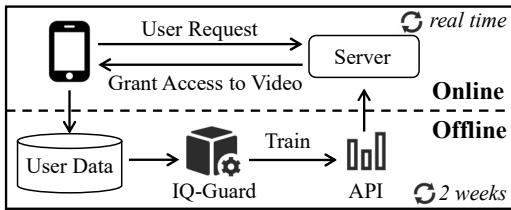
(iii) **iQ-Guard vs. iQ-Guard w/o RLF.** The variant **iQ-Guard w/o RLF** removes the Robust Loss Function, instead using standard cross-entropy (CE) loss. This substitution leads to average decreases of 3.87% in AUC and 5.67% in F1. The performance degradation arises because CE lacks inherent noise resilience, limiting its effectiveness in noisy environments.

4.5 Empirical Study of iQ-Guard (RQ4)

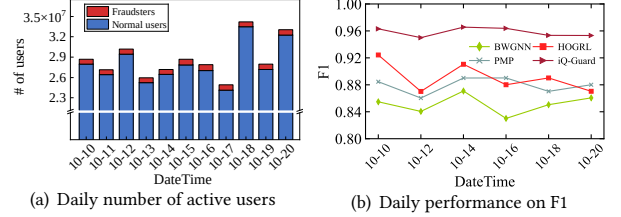
We further conduct an empirical study of **iQ-Guard** to better understand how it addresses **structural noise** and **label noise**. Specifically, we visualize the learned representation distributions and analyze the evolution of label confidence during training. These experiments are designed to provide deeper insights into the underlying mechanisms of iQ-Guard and to explain why it achieves strong robustness in noisy environments. Detailed analyses and visualizations are presented in [Appendix C.6](#).

4.6 Deployment and Application (RQ5)

iQ-Guard has been deployed on the iQIYI platform for five months, supporting daily fraud detection with weekly model updates trained on the previous two weeks of user data. After each update, the model is deployed via an API to perform daily inference and predict

**Figure 5: Architecture of iQ-Guard online deployment.****Table 3: Results on industrial effectiveness and efficiency.**

Methods	T+6↑	SUR↑	CR↓	Training (min)↓	Inference (s)↓
Expert rules	0.0341	0.1028	1.9349	-	-
BWGNN	0.0728	0.1527	1.7651	179	724
PMP	0.0773	0.1957	1.2648	243	1137
HOGRL	0.0847	0.1922	0.9920	228	1008
iQ-Guard	0.1043	0.2593	0.6847	211	983

**Figure 6: Performance on iQIYI's daily fraud detection.**

fraudulent users in real time (Fig. 5). On average, it detects 36 fraud cases per day, preventing an estimated \$1.12 million in financial losses. Feedback from iQIYI has been highly positive, highlighting both the system's accuracy and its operational value.

4.6.1 Quantitative evaluation. To evaluate the business impact and efficiency of iQ-Guard, we conducted a comprehensive quantitative study incorporating both industrial effectiveness metrics and computational efficiency. Specifically, we report performance on three widely adopted business indicators, **T+6**, **SUR**, and **CR**, as well as **training and inference time**.

- **T+6:** The proportion of advertisement-targeted users who make a payment on iQIYI within the next six days.
- **SUR (Start-Up Ratio):** The proportion of advertisement-targeted users who log into iQIYI the following day.
- **CR (Cost Ratio):** The ratio of iQIYI's advertising expenditure to the revenue generated from advertisements. Unlike the other two metrics, a lower CR is preferable.
- **Training and Inference Time:** Computational efficiency measured by the total training time (minutes) on the full dataset and the inference time (seconds) on 500,000 samples.

We evaluate iQ-Guard against iQIYI's expert rules and state-of-the-art GFD baselines, as shown in Table 3. **iQ-Guard consistently achieves the best performance across all evaluation dimensions**, surpassing expert rules by 205.95%, 152.28%, and 64.60% on T+6, SUR, and CR, respectively. While baseline GFD models outperform expert rules, they remain far behind iQ-Guard, highlighting its practical advantage. In addition, iQ-Guard demonstrates superior efficiency: its training time is 13.2% faster than PMP and 7.5% faster than HOGRL, with lower inference latency as well.

Moreover, we have evaluated the daily fraud detection performance of iQ-Guard on iQIYI. As shown in Fig. 6(a), the platform processes tens of millions of visits per day, making reliable detection crucial. Using expert-labeled daily samples, we compared F1 scores (AUC and AP showed similar trends) across methods (Fig. 6(b)). iQ-Guard maintains a stable daily F1 of 0.96 and outperforms others by an average of 6.28%, while competing models fluctuate considerably. These results demonstrate both the robustness and consistency of iQ-Guard in a real-world production environment.

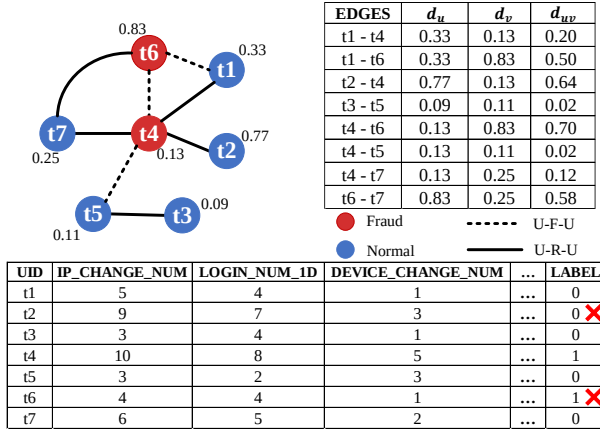


Figure 7: Explanation examples from iQIYI dataset.

4.6.2 Qualitative evaluation. We also analyze representative cases from the iQIYI dataset to illustrate how iQ-Guard improves fraud detection. Fig. 7 presents a multi-relational user graph (t_1 – t_7) with features and labels. Fraudsters often exhibit behaviors such as frequent IP changes and multi-account logins. However, due to the limitations of expert rules, t_2 and t_6 were mislabeled.

iQ-Guard computes both label confidence (d_u) and structural confidence (d_{uv}) for each node. As shown in the top-right table, t_2 and t_6 have the highest JSD values and elevated d_{uv} scores, revealing inconsistencies that point to label noise. This confirms that iQ-Guard’s confidence-based correction mechanism effectively mitigates label errors during training.

4.7 Model Robust Study (RQ6)

We further assess the robustness of iQ-Guard under varying parameter configurations. This evaluation aims to provide a comprehensive understanding of how these factors influence the model’s performance. The detailed results can be found in [Appendix C.7](#).

5 Related Work

A. Graph Fraud Detection

Research on graph fraud detection follows two main strategies:

(i) *Spatial-based Methods.* GNN-based approaches enhance robustness against camouflage and data limitations. A key challenge is the mix of homophily and heterophily caused by fraudsters connecting to legitimate users. Models such as H2-FDetector [44] and SparseGAD [21] design tailored aggregation for such structures, while others mitigate structural and feature noise via reinforcement learning neighbor selection [70] or graph augmentation [63].

Under limited labeled data, earlier work addressed class imbalance [27, 37], while recent methods use unlabeled data in barely-supervised [67] or fully unsupervised settings [40]. Richer contexts are leveraged through temporal modeling [58, 59, 68, 71], multi-relational information [17, 23], and architectural innovations like graph transformers [34] or dynamic grouping aggregation [14]. Adversarial robustness is also pursued via attack-resistant architectures [56] and simulators like MonTi [8].

(ii) *Spectral-based Methods.* These approaches apply graph Fourier transforms to detect fraud via frequency-specific analysis. AM-Net [3] adapts multi-frequency filters to balance smoothing and

anomaly detection, while BWGNN [45] addresses spectral right-shift with localized band-pass filters. SplitGNN [54] reduces heterophily noise by decomposing graphs into homophilic/heterophilic subgraphs with separate filters. Recent works integrate hybrid filtering and invariant learning, e.g., GHRN [19] captures high-frequency inter-class edges via Laplacian operators, SEC-GFD [61] combines mixed-frequency bands with label-guided constraints, and HGIF [42] enhances robustness through edge-aware augmentation, dual-channel contrastive learning, and test-time training.

B. Label Noise Learning

Learning with noisy labels is a critical challenge, especially for graph-based models where the message-passing mechanism can propagate the influence of incorrect labels across the graph. Existing label noise learning can be classified into two categories: (i) *Robust Regularization*, which prevents overfitting to noisy labels by adding regularization terms [11, 24, 41, 52, 74]; (ii) *Sample Separation*, which identifies and separates noisy samples from clean ones, typically using confidence-based filtering techniques [18, 26, 33, 69, 75]. State-of-the-art methods commonly employ the small-loss principle [33], which identifies noisy samples based on their output loss values.

C. Structure Noise Learning

The performance of GNNs is highly dependent on the quality of the graph structure. In fraud detection, this structure is often compromised by noise introduced by fraudsters to evade detection. Existing structural noise learning methods can be broadly categorized into two types: (i) *Graph Rewiring*, which jointly optimizes the graph structure and model parameters to mitigate noisy edges [1, 16, 28, 39, 66, 73]; (ii) *Denoised Graph Construction*, which builds cleaner graphs via link prediction or edge filtering under noisy and sparse settings [12]. The Edge Signal-to-Noise Ratio [13], based on random matrix theory, has been proposed to quantify structural noise and shows strong correlation with GNN performance degradation. A benchmark [72] reveals that rewiring methods vary in robustness across different noise patterns and structural noise further impacts GNN interpretability [30].

6 Conclusion

We propose iQ-Guard, a novel, effective, and noise-resistant framework for graph fraud detection on the iQIYI platform. iQ-Guard simultaneously addresses structural and label noise by integrating two key innovations. First, Dynamic Contrastive Learning tailors data augmentation per sample, maximizing feature utilization while adapting to noisy environments. Second, Structure-Aware Noise-Resistant Training leverages both structural relations and a soft logical perspective to enhance robustness against noise. Extensive experiments on four public datasets confirm its effectiveness, while real-world deployment on iQIYI has led to the detection of 36 fraud cases daily, preventing \$1.12 million in financial losses. Future work will explore handling class imbalance in noisy graph fraud detection to further unveil camouflaged fraudsters.

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A Notations and descriptions

The key notations and descriptions are summarized in Table 4.

Table 4: Notations and descriptions

Notation	Description
$\mathcal{G}$	Graph with nodes, edges, features, and labels
$\mathcal{V}, \mathcal{E}$	Sets of nodes and edges, respectively
$\mathcal{X}, \mathcal{Y}$	Sets of node features and labels, respectively
v, u	Individual nodes in the graph
x_v, y_v	Feature embedding and label for node v
x'_v	Augmented feature embedding for node v
Δ_v	Dynamically scaled noise for data augmentation
d_v	JSD-based disagreement score for node v
p_v, p'_v	Predicted probability for x_v and x'_v
$h_r^{(l)}(v)$	Embedding of node v at layer l under relation r
d_{vu}	Structural confidence score for edge (v, u)
y'_{vu}	Softened label for edge (v, u)
θ, ϕ, ψ	Parameters for the GNN, CL layer, and MLP
ϵ	Perturbation radius for dynamic data augmentation

B Model training

B.1 Training Process of iQ-Guard

The complete implementation process of the iQ-Guard framework is shown in Algorithm 1, including dynamic contrastive learning and structure-aware noise-resistant training.

Algorithm 1: iQ-Guard Training

```

1 Input: a training set  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{X}, \mathcal{Y})$ , a network  $\mathcal{M}(\cdot)$  include
    $\theta, \phi$ , where  $\theta$  represent structure-aware message-passing
   paradigm,  $\phi$  represent dynamic contrastive layer, a three-layer
   network parameterized by  $\psi$ , a radius  $\epsilon$ , a maximum number of
   epochs  $max\_epoch$ .
2  $\phi = \text{WarmUP}(\mathcal{G}, \phi)$  // training only on contrastive layer
3 while  $e < max\_epoch$  do
4   foreach  $v \in \mathcal{V}$  do
5      $p_v = \mathcal{M}(x_v)$ 
6      $d_v$  is calculated through  $p_v$  and  $y_v \in \mathcal{Y}$  by Equation 1
7      $\Delta_v = \delta \odot \text{sgn}(x_v), \delta \in \mathbb{R}^n \sim \mathcal{U}(0, 1 - d_v), \|\Delta_v\|_2 = \epsilon$ 
8      $x'_v = x_v + \Delta_v$ 
9      $p'_v = \mathcal{M}(x'_v)$ 
10     $\mathcal{L}_{cons} = -\sum_{v \in \mathcal{V}} (1 - d_v) y_v \cdot \ln p'_v$ 
11     $\mathcal{L}_{div} = -\sum_{v \in \mathcal{V}} (1 - d_v) q(x_v, x'_v)$ 
12     $\mathcal{L}_{CL} = \mathcal{L}_{cons} + \mathcal{L}_{div}$ 
13    foreach  $v \in \mathcal{V}$  do
14      foreach  $u \in \mathcal{N}_v$  do
15         $d_{vu}$  is calculated through  $d_v$  and  $d_u$  by Equation 14
16         $y'_{vu}$  is calculated by Equation 15
17       $\mathcal{L}_{edge} = -\sum_{v \in \mathcal{V}} \sum_{u \in \mathcal{N}_v} y'_{vu} \ln p_{vu}$ 
18       $\mathcal{L}_{node}$  is calculated by Equation 13
19       $\mathcal{L}_{fraud} = \mathcal{L}_{CL} + \mathcal{L}_{edge} + \mathcal{L}_{node}$ 
20       $\theta, \phi, \psi = \text{Adam}(\mathcal{L}_{fraud}, \theta, \phi, \psi)$ 

```

B.2 Complexity analysis

As shown in Algorithm 1, iQ-Guard include two components: (i) **Dynamic Contrastive Learning** and (ii) **Structure-aware Noise-resistant Training**. These components iterate during each iteration of the process. Compared to the standard training process of

Table 5: Statistics of datasets.

Dataset	#Nodes (Fraud%)	Relation	#Edges	ϕ_r	#Features
Amazon	11,944 (9.5%)	U-P-U	175,608	0.1673	25
		U-S-U	3,566,479	0.0558	
		U-V-U	1,036,737	0.0532	
YelpChi	45,954 (14.5%)	R-U-R	49,315	0.9089	32
		R-S-R	3,402,743	0.1857	
		R-T-R	573,616	0.1764	
T-Finance	39,357 (4.5%)	U-T-U	21,222,543	0.4363	10
T-Social	5,781,065 (3.0%)	U-F-U	73,105,508	0.1003	10
iQIYI	248,214 (2.8%)	U-I-U	7,539,112	0.2136	98
		U-R-U	4,049,302	0.3349	
		U-F-U	3,318,471	0.6131	
		U-M-U	950,327	0.5847	

graph neural networks, this approach may incur higher computational costs. In this section, we provide a complexity analysis.

- **Dynamic Contrastive Learning** has a linear time complexity of $O(N \cdot (C + d))$, where N is the number of nodes, C is the number of classes, and d is the feature dimension. The complexity is derived by considering three stages: (1) **Label Confidence Computation**, where the JSD between predicted and true labels is calculated for each node, resulting in $O(C)$ per node and $O(N \cdot C)$ overall; (2) **Dynamic Data Augmentation**, which involves generating a noise vector for each node and augmenting its feature embedding, with time complexity of $O(d)$ per node and $O(N \cdot d)$ in total; (3) **Contrastive Loss Function**, which includes both CE with complexity $O(N \cdot C)$ and diversity loss with complexity $O(N \cdot d)$, summing to $O(N \cdot (C + d))$. Therefore, the combined time complexity for DCL is $O(N \cdot (C + d))$.
- **Structure-aware Noise-resistant Training** has a total time complexity of $O(LREd^2 + Ed^2 + Vd)$, where E is the number of edges, L is the number of GNN layers, R is the number of relation types. For each of the L layers and R relations, the SNT applies an MLP to each edge’s node-pair embeddings, incurring $O(d^2)$ per edge and totaling $O(LREd^2)$ for all edges. Robust Loss Function (RLF) adds a forward pass for each edge with complexity $O(Ed^2)$, while node classification incurs $O(Vd)$ for processing all nodes.

Overall, the combined complexity per iteration is $O(LREd^2 + Ed^2 + Vd + N \cdot (C + d))$. This dominance indicates that the computational cost is primarily influenced by the number of edges (E), the number of GNN layers (L), the number of relation types (R), and the square of the feature dimension d^2 . While $O(N \cdot (C + d))$ from DCL is significant for large node counts (N), it is often less impactful compared to the more resource-intensive structure-aware operations in SNT. As a result, optimizing L , R , and d while maintaining sufficient model expressiveness is critical for ensuring scalability and computational efficiency in large-scale graph applications.

C Experimental Details

C.1 Datasets

We use four public datasets and one industrial dataset for evaluation. Here, ϕ_r denotes the fraction of edges under relation r whose

incident nodes share the same class label:

$$\phi_r = \frac{|\{(u, v) \mid \mathbf{A}_r[u, v] = 1 \wedge y_u = y_v\}|}{\sum \mathbf{A}_r[u, v]}, \quad (18)$$

where $\mathbf{A}_r$ is the adjacency matrix under relation r , and y_u, y_v are the labels of nodes $u, v \in \mathcal{V}$. A higher ϕ_r indicates stronger homophily for that relation.

- **Amazon**: Reviews from the musical instruments domain with three relations: U-P-U (same product), U-S-U (same rating within one week), and U-V-U (top 5% text similarity by TF-IDF).
- **YelpChi**: Hotel and restaurant reviews with three relations: R-U-R (same user), R-S-R (same product and star), and R-T-R (same product in the same month).
- **T-Finance**: A transaction network where nodes are accounts and edges are transactions; anomalies are labeled by experts.
- **T-Social**: A social network where two users are connected if they maintain a friendship for over three months on a large app.
- **iQIYI**: An industrial dataset from the video platform, containing noisy labels and four relations: U-I-U (same IP), U-R-U (same review), U-F-U (follow), and U-M-U (same MAC).

C.2 Baseline Descriptions

In this section, we will introduce the baselines used in more detail.

- **GCN** [31], **GAT** [47], and **GraphSAGE** [5] represent general GNN models for homogeneous or heterogeneous graphs.
- **NRGNN** [11] and **GraphCleaner** [36] represent general GNN models with the label noise problem.
- **GraphMAE** [25] and **GraphACL** [60] are all state-of-the-art models that leverage contrastive learning or data augmentation for node classification tasks on graphs.
- **BWGNN** [45] detects fraudsters by capturing low and high-frequency signals and adaptively combining signals at different frequencies.
- **SparseGAD** [21] improves detection by unveiling covert dependencies that coordinate with label consistency between node pairs.
- **GAGA** [50] is the graph fraud detection method that leverages Transformer to generate distinct neighborhood information, sharpening fraudulent activity detection semantics.
- **ConsisGAD** [6] takes advantage of the variance in homophily distribution between normal and anomalous nodes to craft a simplified GNN backbone, enhancing its capability to distinguish effectively between these two classes.
- **PMP** [76] distinguishes neighbors with different labels during message passing, using node-specific aggregation functions to adaptively handle both homophilic and heterophilic neighbors, improving model performance on mixed-label graphs.
- **HOGRL** [77] is a high-order graph representation learning model for credit card fraud detection that addresses the over-smoothing issue in multi-layer aggregation.
- **RG TAN** [58], a gated temporal attention network is employed for node message-passing to catch fraudulent representations.

C.3 Evaluation Metrics

We evaluate model performance using the Area Under the ROC Curve (AUC), the Average Precision (AP), and the macro-averaged F1 score (F1):

Table 6: AUC and F1 of iQ-Guard and its variants without key modules. The highest scores are highlighted in bold.

Methods	T-Finance		T-Social	
	AUC	F1	AUC	F1
iQ-Guard	0.9380	0.8836	0.9400	0.8729
iQ-Guard w/o DCL	0.7826	0.8015	0.8341	0.7449
iQ-Guard w/o DCL (ND)	0.8011	0.7927	0.8412	0.7633
iQ-Guard w/o DCL (ED)	0.8048	0.8112	0.8537	0.7820
iQ-Guard w/o DCL (RW)	0.7977	0.7936	0.8444	0.7729
iQ-Guard w/o SMP (GCN)	0.6884	0.6916	0.7622	0.7394
iQ-Guard w/o SMP (PMP)	0.8957	0.8115	0.8947	0.8441
iQ-Guard w/o SMP (HOGRL)	0.9187	0.8584	0.9118	0.8563
iQ-Guard w/o RLF	0.8217	0.7930	0.8031	0.7705

- **AUC:** The area under the ROC curve, reflecting the model’s ability to distinguish between positive and negative samples.

$$AUC = \int_0^1 TPR(f) dFPR(f). \quad (19)$$

- **AP:** The area under the Precision–Recall curve, emphasizing the ranking of positive samples.

$$AP = \int_0^1 \text{Precision}(r) dr, \quad (20)$$

where $\text{Precision}(r)$ is the precision at rank r .

- **F1:** The average F1 score across all classes, balancing precision P_i and recall R_i .

$$F1 = \frac{1}{l} \sum_{i=1}^l \frac{2P_i R_i}{P_i + R_i}. \quad (21)$$

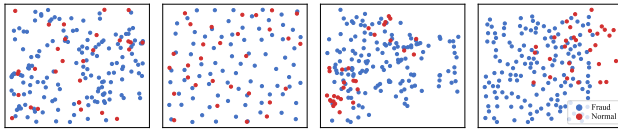
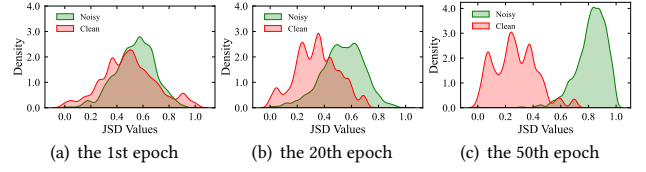
Since datasets may exhibit highly skewed label distributions, AUC alone can be overly optimistic. To obtain a more comprehensive evaluation, we also report AP and F1. All metrics range from 0 to 1, with higher values indicating better performance.

C.4 Implementation Details

All experiments of iQ-Guard are conducted using PyTorch in Python 3.8 and run on an NVIDIA A40 GPU. The dataset is split into training/validation/testing with a ratio of 0.4/0.1/0.5. We set the radius $\epsilon = 0.75$ for all experiments. Baseline hyperparameters follow the settings in the original papers.

C.5 Ablation Study

Table 6 performs a full ablation study on T-Finance and T-Social. The observations align with the discussion presented in the main paper. The results confirm that each key module plays a crucial role in the performance of iQ-Guard.

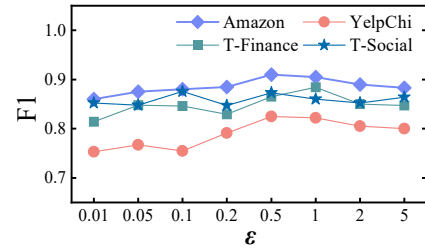
**Figure 8: Representation distributions on Amazon.****Figure 9: Performance of JSD values over epochs.**

C.6 Empirical Study

We visualize the learned representation distributions and track the evolution of label confidence throughout training to gain a deeper understanding of iQ-Guard’s underlying mechanisms and its robustness in noisy environments.

i) Representation Distribution. We apply DCL, HOGRL, and iQ-Guard to the Amazon dataset with a 20% noisy ratio and visualize the final output using t-SNE [46]. As illustrated in Fig. 8, the original distribution is irregular, making it difficult to distinguish between fraudsters and normal users. After applying DCL, the distribution aligns with expectations, as contrastive learning aims to evenly distribute the entire representation space [7]. Next, we present the results of the most competitive methods. After applying HOGRL, the Amazon data exhibits some clustering of fraudsters and normal users, but many normal users are still incorrectly clustered near fraudsters. Finally, the data processed by iQ-Guard demonstrates better clustering, with a clear distinction between fraudsters and normal users.

ii) Label Confidence Distribution. We present the distribution of Jensen-Shannon Divergence (JSD) values at different stages of iQ-Guard to illustrate the changes in label confidence. As shown in Fig. 9, the four distributions from the Amazon dataset are visualized using Gaussian Kernel Density Estimation. As observed, as training progresses, the distinction between the JSD distributions of noisy and clean samples becomes increasingly apparent. The JSD values of noisy samples shift to the right, while those of clean labels shift to the left. This indicates that iQ-Guard achieves noise resistance, remaining unaffected by noisy during training.

**Figure 10: Performance on F1 under different radius ϵ**

C.7 Parameter Study

We investigate the impact of ϵ in iQ-Guard. Only F1 is shown here, AUC and AP have the same change trend. We think a larger ϵ leads to a more uniform distribution that can help debias. However, when it is too large, fraud detection will be hindered because the high similarity between connected nodes cannot be reflected by an over-uniform distribution. As shown in Fig. 10, the shapes of the curves are as expected. On all the datasets, when ϵ is near 0.75, iQ-Guard achieves the best performance.